



Implementation and Monitoring MLOps Solution

LECTURE

Type of Model Monitoring



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The lecture **Type of Model Monitoring** explains different monitoring types, drift categories, and scenarios that trigger ML model retraining.

Type of Model Monitoring

Business Requirements

- Ensuring the ML solution **aligns with and fulfills specific business objectives**.
- Regular assessments to ensure **continued relevance** to evolving business needs.

Model Performance

- Tracking the **accuracy and efficiency** of the model over time.
- Detecting and addressing types of **drift or degradation**.

Resource Utilization

- Ensuring **efficient resource utilization** within the ML infrastructure.
- **Compliance** with Service Level Agreements (SLAs) for system performance and availability.

Traceability

- Facilitating **audit trails** for troubleshooting, regulatory compliance, and model improvement.
- **Tracking data lineage** to understand the origin, movement, and transformation of data.



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MLOps practices must align with business objectives by ensuring models directly support strategic goals and remain relevant through regular evaluations. Continuous monitoring tracks accuracy, efficiency, and drift to maintain reliable performance, while efficient resource utilization ensures cost control and SLA compliance. Strong traceability is achieved through audit trails for compliance and troubleshooting, along with detailed data lineage to provide transparency, accountability, and confidence in the model lifecycle.

Four Types of Drift

Data Drift:

- Occurs when the **statistical properties** of the input data **change over time**.
- Can impact the model's quality by **introducing inconsistencies** in the data patterns.

Concept Drift:

- Happens when the **relationship** between input features and the **target variable changes**.
- Forces models to adapt to **new patterns** to stay relevant.

Model Quality Drift:

- Reflects a decrease in the model's predictive performance over time.
- Can be **detected through worsening metrics like accuracy, precision, recall, or F1 score**.

Bias Drift:

- Involves shifts in model outcomes that could lead to unfair treatment of certain groups.
- Monitoring for bias drift is **crucial to maintain fairness and ethical standards** in model predictions.



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Data quality drift occurs when the statistical distribution of incoming data differs from the training data, such as shifts in distribution, noisy inputs, or missing values.

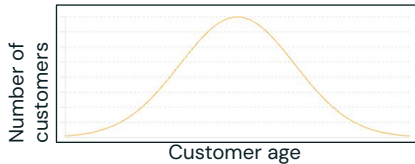
Concept drift arises when relationships between features and targets change, like evolving fraud detection patterns. Model quality drift reflects declining predictive accuracy over time, measured through metrics such as MSE, R^2 , accuracy, precision, or recall, often seen in recommendation systems as user behavior evolves. Bias drift happens when group distributions in production differ from training, for example, the appearance of a new group or underrepresentation of certain customers, leading to degraded model performance.

Illustrating Data and Model Drift

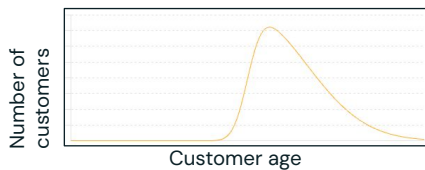
Visualizing Changes in Data and Model Performance Over Time

Data Drift

Training data distribution:



Production data distribution:



Model Drift

Right After Deployment:

Predicted Label	Positive	Negative
	80	5
True Label	Positive	10
	Negative	85

After a Period of Time:

Predicted Label	Positive	Negative
	50	35
True Label	Positive	10
	Negative	85



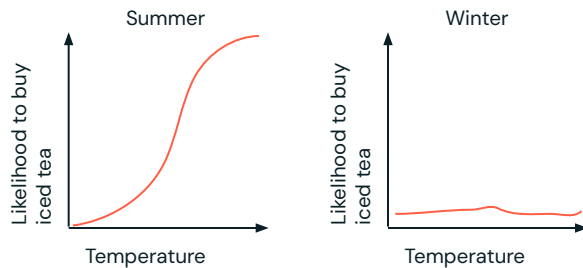
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A data drift scenario can be seen with QuickFashions, an online retail company, where a recommendation model trained on uniformly distributed purchase history data encountered seasonal shifts in production data. As purchases became skewed toward categories like winter wear, the system continued recommending summer clothes during peak winter demand, reducing engagement, sales, and customer satisfaction. A model drift scenario occurs with GreenEnergy, which classifies solar panel maintenance as 'urgent' or 'routine.' Precision-recall metrics are critical to avoid false alarms and missed issues. A precision drop signals too many false positives, while a recall drop means urgent needs are overlooked—both leading to costly downtimes and reduced service reliability.

Illustrating Concept and Bias Drift

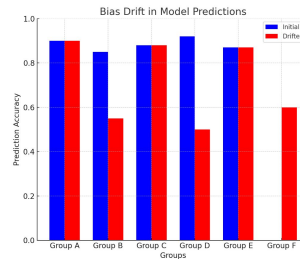
Visualizing Changes in Concept and Bias Over Time

Concept Drift

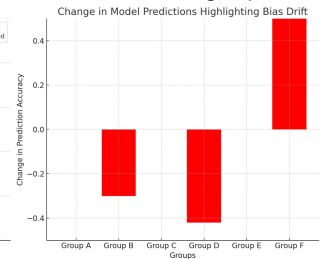


Bias Drift

Change in Group Distribution & New Group



Prediction accuracy varies across different groups



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A concept drift scenario occurs at AgriSmart, an agricultural technology firm, where a crop yield model trained on soil quality, moisture, nutrients, and pest presence faced new production data. The introduction of a “genetic modifications” feature and the emergence of an invasive pest changed the factors influencing yields. As a result, the model underestimated these effects, reducing forecast accuracy and impairing crop management and resource allocation decisions. A bias drift scenario occurs at GlobalMusic, an international streaming service, where new regional markets with distinct musical preferences altered user demographics. The recommendation engine began under-recommending regional music to these new groups, lowering engagement and subscription renewals in the affected markets.

Data Drift Scenario:

Event: Introduction of a New Product Line

Scenario: The company decides to introduce a new line of eco-friendly products, heavily promoting them through social media and email campaigns. This new product line attracts a different demographic compared to the existing customer base.



Changes in Data:

- Demographic Shift
- Browsing Behavior
- Historical Sales Data
- Promotional Activities



Impact of Data Drift:

- **Prediction Accuracy:** Training data no longer represents the current customer behavior and product offerings.
- **Sales Forecasting:** The sales of eco-friendly products is underestimated and the sales of other products is overestimated.



Addressing Data Drift:

- Collect New Data
- Retrain the Model
- Monitor Continuously

Note: The scenario does overlap with concept drift.



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- **Demographic Shift:** The new eco-friendly products attract younger, environmentally conscious customers who were not previously part of the regular customer base.
- **Browsing Behavior:** The new customers spend more time reading product descriptions and reviews for the eco-friendly products compared to other products.
- **Historical Sales Data:** The historical sales data, which the model heavily relies on, does not include information about the newly introduced product line.
- **Promotional Activities:** The impact of promotions changes due to targeted marketing for the eco-friendly product line.
 - Social media campaigns become a more significant driver of sales compared to email promotions.

ML Model Retraining Triggers

- **Scheduled Retraining:**
 - Databricks recommends starting with scheduled, periodic retraining and moving to triggered retraining when needed.
- **Data Changes:**
 - Changes in the data can either explicitly trigger a retraining job or it can be automated if data drift is detected.
- **Model Code Changes:**
 - Retraining can be triggered by changes in the model code, often due to concept drift or other factors that necessitate an update in the model.
- **Model Configuration Changes:**
 - Alterations in the model configuration can also initiate a retraining job.
- **Monitoring and Alerts:**
 - Jobs can monitor data and model drift, and Databricks SQL dashboards can display status and send alerts



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- **Scheduled Retraining Example:** Retraining a recommendation system every month to include the latest user behavior data.
- **Data Changes Example:** A sudden increase in sales during a holiday season requires the sales prediction model to be retrained.
- **Model Code Changes Example:** Introducing a new feature to the model requires retraining to incorporate this feature effectively.
- **Model Configuration Changes Example:** Tuning hyperparameters like learning rate or batch size to improve model performance.
- **Monitoring and Alerts Example:** Retraining a fraud detection model when an increase in false positives is detected (Model Quality Drift)



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